ÖZYEĞİN UNIVERSITY

Fall 2021

MATH 104 Midterm

Name: .....

Number: .....

- The exam will be 100 minutes long and consists of 9 questions.
- **Be sure to write full name and student number and sign each answer sheet.**
- Calculators and other electronic devices such as smart phones are not allowed during the exam.
- **Answers given without supporting work will receive no credit.**
- If one of the proctors suspects you of communicating with one of your fellow students or other ways of cheating, you will be immediately removed from the the exam room. This incident will be reported to the Dean's office.

Honor Code: I pledge my honor that I **did not give or receive any unauthorized assistance** in this work.

Signed:.....

Question	Max. Points	Score
1	10	
2	15	
3	15	
4	16	
5	8	
6	10	
7	10	
8	5	
9	11	
Total	100	

Q1) *(Infinite Series)* Which of the following series converge, and which diverge? If a series converges, find its sum.

$$i)\ \sum a_n = \sum_{n=2}^{\infty}\left(\frac{1}{n} - \frac{1}{n-1}\right)$$

n^{th} partial sum of the series

$S_n = a_2 + a_3 + \cdots + a_n$

$$= \left(\cancel{\tfrac{1}{2}} - \tfrac{1}{1}\right) + \left(\cancel{\tfrac{1}{3}} - \cancel{\tfrac{1}{2}}\right) + \left(\cancel{\tfrac{1}{4}} - \cancel{\tfrac{1}{3}}\right) + \cdots + \left(\cancel{\tfrac{1}{n-1}} - \cancel{\tfrac{1}{n-2}}\right) + \left(\tfrac{1}{n} - \cancel{\tfrac{1}{n-1}}\right)$$

$S_n = -1 + \frac{1}{n}$

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty}\left(-1 + \frac{1}{n}\right) = -1$$

$$\sum_{n=2}^{\infty}\left(\frac{1}{n} - \frac{1}{n-1}\right) = -1$$ converges.

$$ii)\ \sum a_n = \sum_{n=3}^{\infty}\left(1 - \frac{2}{n}\right)^n$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(1 - \frac{2}{n}\right)^n = e^{-2} \neq 0$$

Since the limit is different from zero,

the series $\sum_{n=3}^{\infty}\left(1 - \frac{2}{n}\right)^n$ diverges by the nth term test

Q2) *(Comparison, Ratio and Root Tests)* Use any method to determine if each series converges or diverges. Give reasons for your answer.

$$i)\ \sum a_n = \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Apply the Ratio Test

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \frac{n!}{2^n} = \frac{2}{n+1} \to 0 = \rho < 1.$$

The series $\sum \frac{2^n}{n!}$ converges by the Ratio Test

$$ii)\ \sum a_n = \sum_{n=1}^{\infty} \frac{(2n)^n}{3^{n+1}},$$

Apply the Root Tst

$$\sqrt[n]{a_n} = \left[\frac{(2n)^n}{3^n 3}\right]^{\frac{1}{n}} = \frac{2n}{3\sqrt[n]{3}} \to \infty > 1$$

The series $\sum \frac{(2n)^n}{3^{n+1}}$ diverges by the Root Test.

$$iii)\ \sum a_n = \sum_{n=2}^{\infty} \frac{2n-1}{n^2-1}$$

Apply the LCT:

$$a_n = \frac{2n-1}{n^2-1} \sim 2\frac{1}{n} = b_n \text{ for large } n.$$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim \frac{2n-1}{n^2-1} \frac{n}{2} \to 1 > 0 \text{ and}$$

By the LCT both series $\sum a_n$ and $\sum b_n$ diverge since the harmonic series $\sum \frac{1}{n}$ diverges

Q3) *(Power Series)* Given that $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n} = \sum a_n$, *i)* find the series' radius of convergence and interval of convergence. For what values of x does the series converges *ii)* absolutely, *iii)* conditionally? *iv)* Use the term-by-term differentiation to find the sum of the series in its interval of convergence

- Apply the Ratio Test to $\sum |a_n|$: $\frac{|x-a|}{R} < 1$

$$\frac{|a_{n+1}|}{|a_n|} = \frac{|x-2|^{n+1}}{n+1} \frac{n}{|x-2|^n} = \underbrace{\frac{n}{n+1}}_{\to 1} |x-2| \to |x-1| = \rho; \quad R=1$$

For $\rho = |x-1| < 1$ the PS converges absolutely: $1 < x < 3$

- Convergence at endpoints:

$x=1$: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ Alternating HS converges by the Alternating series Test, PS converges conditionally at $x=1$

$x=3$: $\sum_{n=1}^{\infty} \frac{1}{n}$ Harmonic series diverges.

- Interval of convergence: $1 \le x < 3$

- $f(x) = \sum_{n=1}^{\infty} \frac{(x-2)^n}{n} = ?$ $1 < x < 3$ $\quad f(2) = 0$

Term-by-term differentiation gives

$$f'(x) = \sum_{n=1}^{\infty} (x-2)^{n-1} \quad (\text{Geometric series conv. If } |x-2| < 1)$$

$$f'(x) = \frac{1}{1-(x-2)} = \frac{1}{\underbrace{3-x}_{>0}} \quad \text{for } 1 < x < 3$$

$$\int f'(x)\,dx = \int \frac{dx}{3-x} \Rightarrow f(x) = -\ln(3-x) + C \quad \text{with } f(2) = 0$$

$$f(2) = 0 \Rightarrow C = 0$$

Sum of the series: $f(x) = \sum_{n=1}^{\infty} \frac{(x-2)^n}{n} = -\ln(3-x), \; 1 < x < 3$

Q4) *(Maclaurin Series, Taylor Polynomials)* The Maclaurin series generated by $f(x)$ is given by $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$. *i)* Find the Maclaurin series generated by $f(x) = \ln(2-x)$. *ii)* Write also the Taylor polynomials of orders 1 and 2 generated by $f(x) = \ln(2-x)$ at $a = 0$.

$f(x) = \ln(2-x)$, $f'(x) = -(2-x)^{-1}$ $\quad f''(x) = -(2-x)^{-2}$

$f'''(x) = -2(2-x)^{-3}$ $\quad f^{(4)}(x) = -2 \cdot 3 (2-x)^{-4}$

$f^{(n)}(x) = -(n-1)!\,(2-x)^{-n}$ $\quad n \geq 1$

$f(0) = \ln 2$ $\quad f^{n}(0) = -(n-1)!\,2^{-n}$

Maclaurin series generated by $f(x) = \ln(2-x)$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \ln 2 + \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$= \ln 2 - \sum_{n=1}^{\infty} \frac{x^n}{n 2^n}$$

$$= \ln 2 - \frac{x}{2} - \frac{x^2}{8} + \cdots$$

Taylor polynomials:

$P_1(x) = \ln 2 - \frac{x}{2}$

$P_2(x) = \ln 2 - \frac{x}{2} - \frac{x^2}{8}$

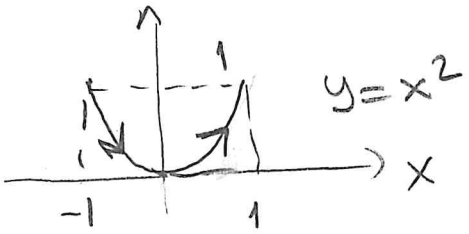
Q5) *(Parametric Curves)* Given the parametric equations and parameter interval for the motion of a particle in the $xy-$plane

$$x = -\cos t, \quad y = \cos^2 t, \quad 0 \le t \le \pi,$$

identify the particle's path by finding a Cartesian equation for it. Graph the Cartesian equation and indicate the direction of the motion and the portion traced by the particle.

$\cos t = -x \Rightarrow y = (-x)^2$ $\qquad t=0 \Rightarrow x=-1$

Cartesian eq. $y = x^2$, $-1 \le x \le 1$ $\qquad t=\pi \Rightarrow x=1$



Q6) *(Polar Coordinates) i)* Find the polar coordinates, $0 \le \theta \le 2\pi$ and $r \le 0$, of the point $(-1, 1)$ given in Cartesian coordinates.

ii) Replace the polar equation $r = -9\sin\theta$ with an equivalent Cartesian equation. Then describe the graph.

i) $(x,y) = (-1,1)$ $\qquad r = \sqrt{x^2+y^2} = \sqrt{2} > 0$

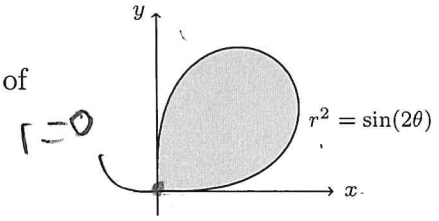
$\tan\theta = \frac{1}{-1} = -1 \Rightarrow \theta = \frac{3\pi}{4}$ (2nd quadrant)

$(r,\theta) = (\sqrt{2}, \frac{3\pi}{4}) = (-\sqrt{2}, \frac{3\pi}{4}+\pi) = (-\sqrt{2}, \frac{7\pi}{4}) = (-r, \theta+\pi)$

ii) $r^2 = -9r\sin\theta \Rightarrow x^2+y^2+9y=0 \Rightarrow x^2 + (y+\frac{9}{2})^2 = \frac{81}{4}$

The circle of radius $\frac{9}{2}$ with center $(0, -\frac{9}{2})$

Q7) *(Areas in Polar Coordinates)* Find the area of the shaded region shown on the right.



$A = \frac{1}{2}\int_\alpha^\beta r^2(\theta)\,d\theta = ?$ $\qquad$ For $r=0$, $\sin 2\theta = 0 \Rightarrow \theta = 0, \frac{\pi}{2}$

$A = \frac{1}{2}\int_0^{\pi/2} (\sin 2\theta)\,d\theta = -\frac{1}{4}\cos 2\theta \Big|_0^{\frac{\pi}{2}} = -\frac{1}{4}(-1-1) = \frac{1}{2}$

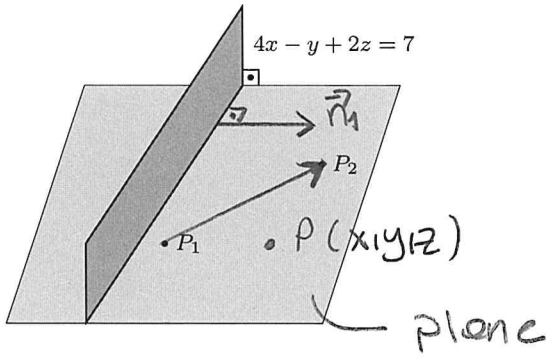
Q8) *(Parametric Equations for a Line in Space)* Find parametric equations for the line through the point $(2, 0, -2)$ parallel to the vector $\boldsymbol{v} = \boldsymbol{i} + \boldsymbol{k}$.

$L: \quad x = x_0 + tv_1 \qquad y = y_0 + tv_2 \qquad z = z_0 + tv_3$

$x = 2 + t \qquad y = 0 + t.0 \qquad z = -2 + t$

$y = 0$

Q9) *(Planes)* Find a plane through the points $P_1(2, -2, 1)$, $P_2(2, 1, 0)$ and perpendicular to the plane $4x - y + 2z = 7$.



- Normal vector $\vec{n}_1$ to $4x - y + 2z = 7$:

  $\vec{n}_1 = \langle 4, -1, 2 \rangle$ is in the plane

- $\overrightarrow{P_1P_2} = \langle 0, 3, -1 \rangle$ is in the plane.

- Normal vector $\vec{n}$ to the plane:

$$\vec{n} = \overrightarrow{P_1P_2} \times \vec{n}_1 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 3 & -1 \\ 4 & -1 & 2 \end{vmatrix} = \vec{i}\,(6-1) - \vec{j}\,(0+4) + \vec{k}\,(0-12)$$

$$\vec{n} = 5\vec{i} - 4\vec{j} - 12\vec{k}$$

- An eq. for the plane through $P_2(2,1,0)$ normal to $\vec{n} = 5\vec{i} - 4\vec{j} - 12\vec{k}$ is $\overrightarrow{PP_2} \cdot \vec{n} = 0$

$$5(x-2) - 4(y-1) - 12(z-0) = 0$$

$$5x - 4y - 12z = 6$$